Kapasa Makasa University

Information Communication Technology Department
(Cyber Security)

Wireless Security

(CYS331)

Lecture 3

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Wireless Personal Area Network

"WPANs"

IEEE definition of WPAN

Wireless personal area networks (WPANs) are used to convey information over short distances among a private, intimate group of participant devices.

Unlike a wireless local area network (WLAN), a connection made through a WPAN involves little or no infrastructure or direct connectivity to the world outside the link. This allows small, power-efficient, inexpensive solutions to be implemented for a wide range of devices.

WPANs

1. Bluetooth
2. Zigbee
3. Ultra Wide Band (UWB)

What is Bluetooth?

- A **cable-replacement** technology that can be used to connect almost any device to any other device
- Radio interface enabling electronic devices to communicate wirelessly via short range (10 meters) ad-hoc radio connections
- a standard for a **small , cheap radio chip to be plugged into computers, printers, mobile phones, etc**

Products

- Notebook PCs & desktop computers
- Printers
- PDAs
- Other handheld devices
- Cell phones
- Wireless periperals:
  - Headsets
  - Cameras
- Access Points
- CD Player
- TV/VCR/DVD
- Telephone Answering Devices
- Cordless Phones
- Cars

What is Bluetooth?

- Uses the radio range of 2.45 GHz
- Theoretical maximum bandwidth is 1 Mb/s
- Several Bluetooth devices can form an ad hoc network called a “piconet”
  - In a piconet one device acts as a master (sets frequency hopping behavior) and the others as slaves
  - Example: A conference room with many laptops wishing to communicate with each other

History

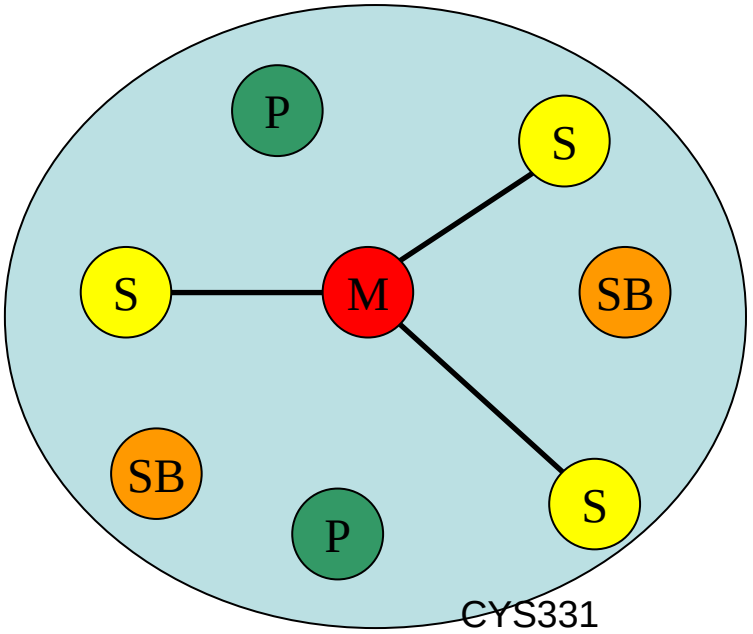
- Harald Bluetooth : 10th century Danish King, managed to unite Denmark and Norway
- Bluetooth SIG (Special Interest Group) :
  - Founded in 1998 by : Ericsson, Intel, IBM, Toshiba and Nokia
  - Currently more than 2500 adopter companies
  - Created in order to promote, shape an define the specification and position Bluetooth in the market place : Bluetooth 2.1

Bluetooth Architecture

- Piconet
  - Each piconet has one master and up to 7 simultaneous slaves
    - **Master** : device that initiates a data exchange.
    - **Slave** : device that responds to the master
- Scatternet
  - Linking of multiple piconets through the master or slave devices
  - Bluetooth devices have point-to-multipoint capability to engage in Scatternet communication.

Piconet

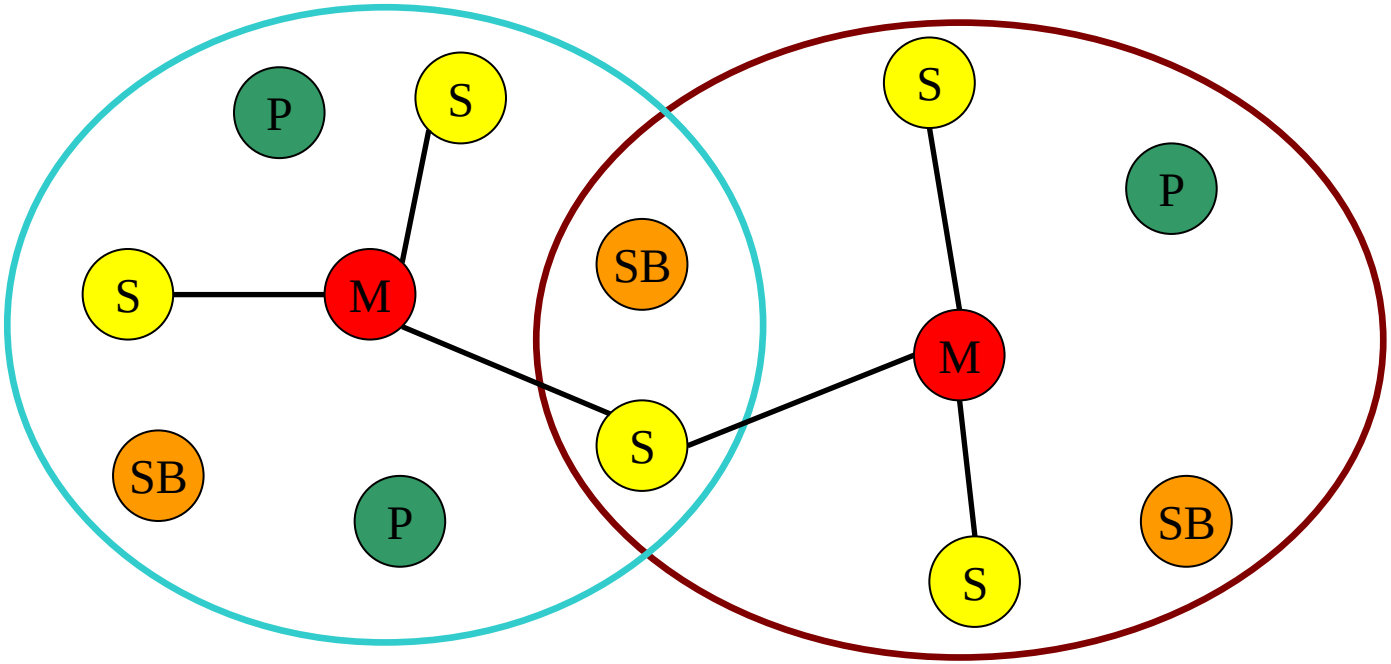
- All devices in a piconet hop together
  - Master gives slaves its clock and device ID
- Non-piconet devices are in standby



M=Master P=Parked
S=Slave SB=Standby

Scatternet

- Devices can be slave in one piconet and master of another



Physical links

- Between master and slave(s), different types of links can be established. Two link types have been defined:
  - Synchronous Connection-Oriented (SCO) link
  - Asynchronous Connection-Less (ACL) link

Physical links

- Synchronous Connection Oriented (SCO)
  - Support symmetrical, circuit-switched, point-to-point connections
  - Typically used for voice traffic.
  - Data rate is 64 kbit/s.
- Asynchronous Connection-Less (ACL)
  - Support symmetrical and asymmetrical, packet-switched, point-to-multipoint connections.
  - Typically used for data transmission .
  - Up to 433.9 kbit/s in symmetric or 723.2/57.6 kbit/s in asymmetric

Bluetooth Security

- The following are the three basic security services specified in the Bluetooth standard:
  - **Authentication**
    - verifying the identity of communicating devices. User authentication is not provided natively by Bluetooth.
  - **Confidentiality**
    - preventing information compromise caused by eavesdropping by ensuring that only authorized devices can access and view data.
  - **Authorization**
    - allowing the control of resources by ensuring that a device is authorized to use a service before permitting it to do so.

ZigBee-Introduction

- ZigBee is a technological standard designed for control and sensor networks
- Based on the IEEE 802.15.4 Standard
- Created by the ZigBee Alliance

ZigBee-Introduction Cont..

- Operates in Personal Area Networks (PAN′s) and device-to-device networks
- Connectivity between small packet devices
- Control of lights, switches, thermostats, appliances, etc.

ZigBee-History

- Developement started 1998, when many enginereers realized that WiFi and Bluetooth were going to be unsuitable for many applications

- IEEE 802.15.4 standard was completed in May 2003

ZigBee Alliance

- Organization defining global standards for reliable, cost-effective, low power wireless applications
- A consortium of end users and solution providers, primarily responsible for the development of the 802.15.4 standard
- Developing applications and network capability utilizing the 802.15.4 packet delivery mechanism

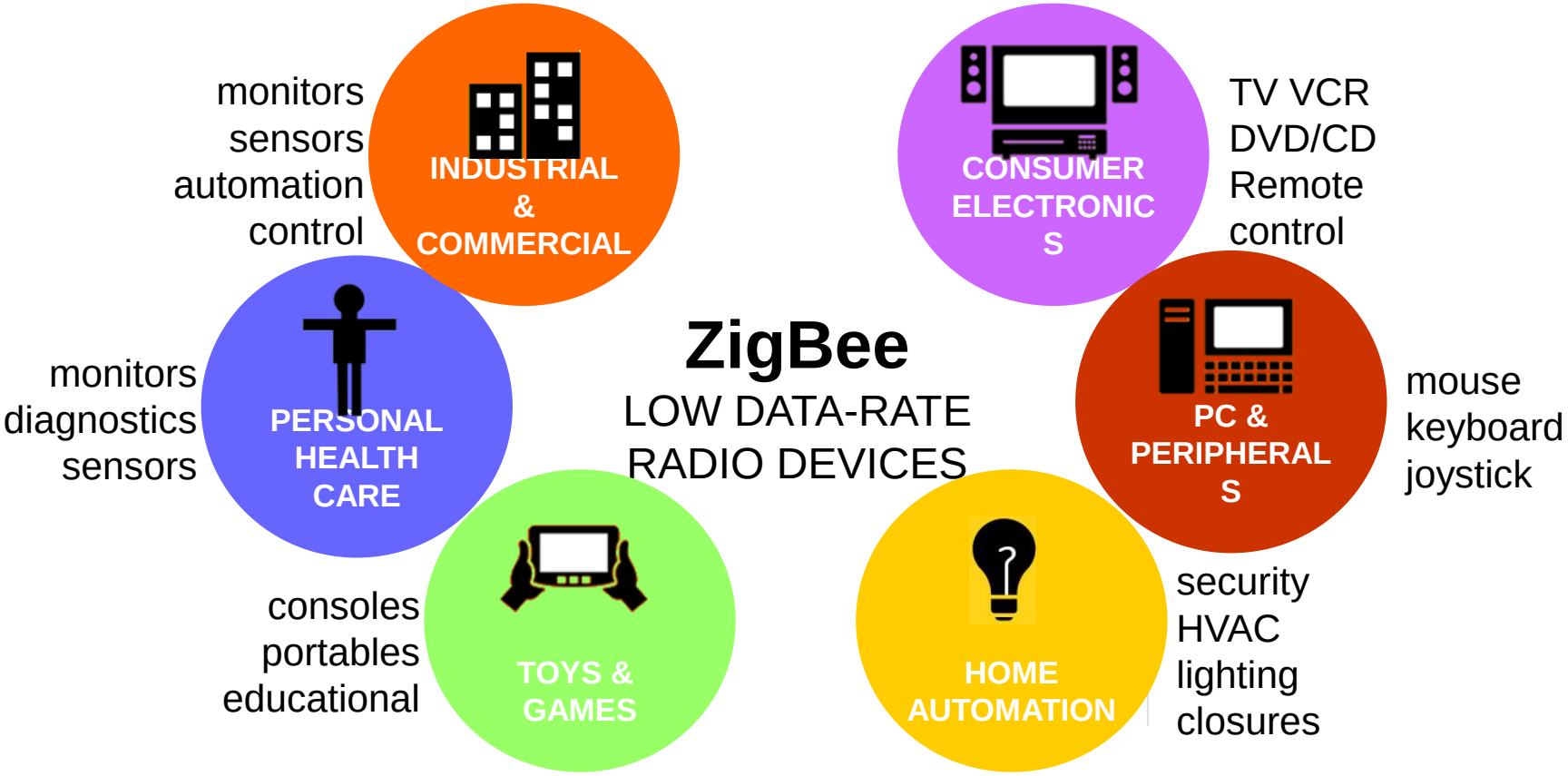
ZigBee-Characteristics

- Low cost
- Low power consumption
- Low data rate
- Relatively short transmission range
- Scalability
- Reliability
- Flexible protocol design suitable for many applications

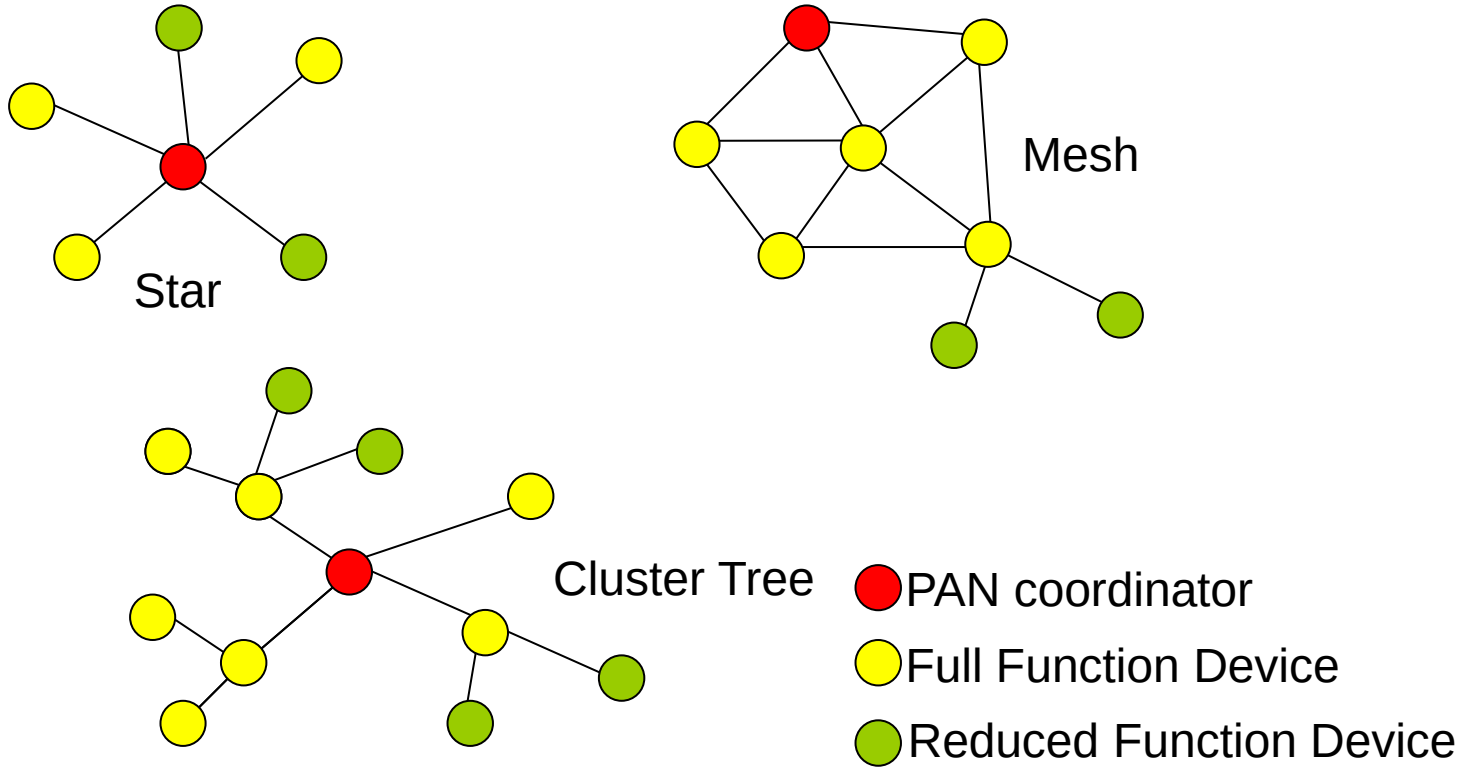
ZigBee-Security

- Encryption specified for MAC, Network and APS layers
- Encryprion/Authentication mode CCM(CTR +CBC-MAC)
  - CTR is a counter based encryption mode
  - CBC-MAC provides data integrity
- All security is based on 128bit key and AES-128 block encryption method

ZigBee-Applications

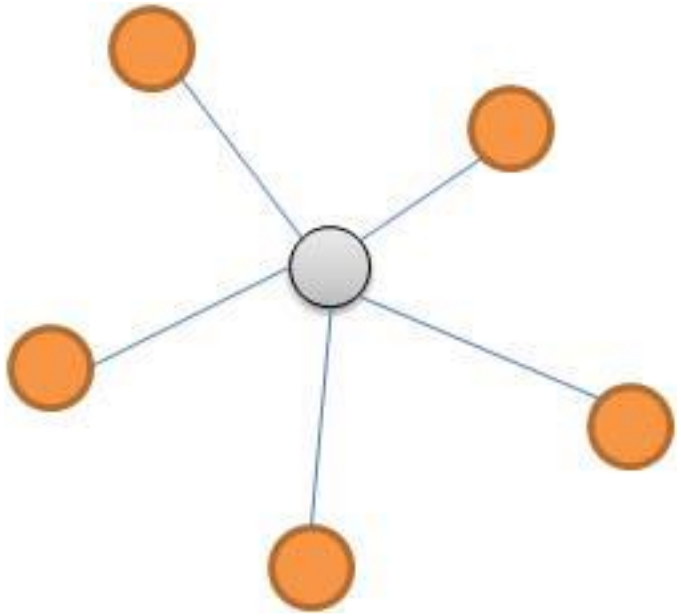


ZigBee Network Topologies



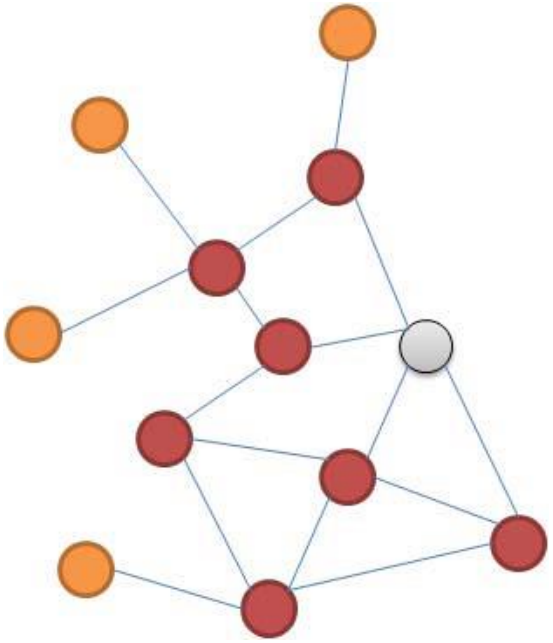
ZigBee Network Topologies

- Star Topology
  - Advantage
    - Easy to synchronize
    - Low latency
  - Disadvantage
    - Small scale



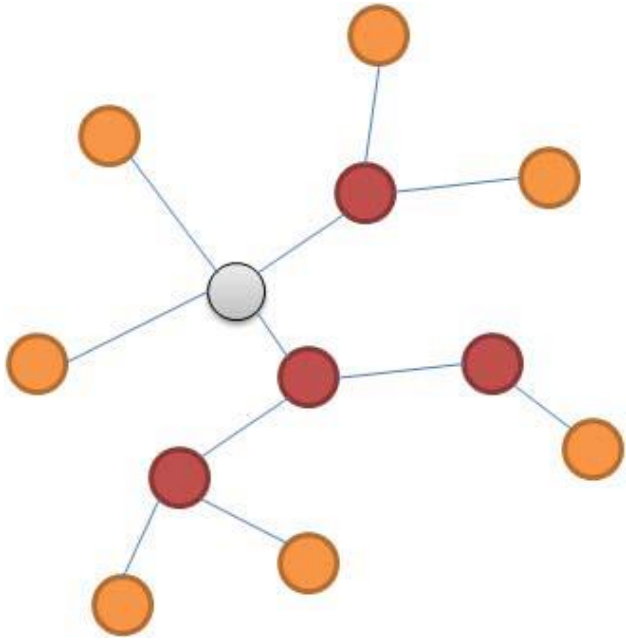
ZigBee Network Topologies

- Mesh Topology
  - Advantage
    - Robust multihop communication
    - Network is more flexible
    - Lower latency
  - Disadvantage
    - Route discovery is costly
    - Needs storage for routing table



ZigBee Network Topologies

- Cluster Tree
  - Advantage
    - Low routing cost
    - Allow multihop communication
  - Disadvantage
    - Route reconstruction is costly
    - Latency may be quite long



ZigBee and Bluetooth Comparison

- *Optimized for different applications*
  - ZigBee
    - Smaller packets over large network
    - Mostly Static networks with many, infrequently used devices
    - Home automation, toys, remote controls, etc.
  - Bluetooth
    - Larger packets over small network
    - Ad-hoc networks
    - File transfer
    - Screen graphics, pictures, handsfree audio, Mobile phones, headsets, PDAs, etc.

ZigBee and Bluetooth Comparison

Feature(s)	Bluetooth	ZigBee
Power Profile	days	years
Complexity	complex	Simple
Nodes/Master	7	64000
Latency	10 seconds	30 ms – 1s
Range	10m	70m ~ 300m
Extendibility	no	Yes
Data Rate	1 Mbps	250 Kbps
Security	64bit, 128bit	128bit AES and Application Layer

Ultrawideband

Contents

Introduction

- Technology for short range Wireless radio communications
- Bandwidth more than 25% of a center frequency or more than 1.5 GHz
- Different from conventional narrowband and wideband systems

Why Ultrawideband

- Crunch in Radio Frequency spectrum availability
- Variations in RF spectrum from one country to next
- Devices using RF Spectrum are more complex, cost more, and consume more power

Why is UWB unique?

- Simultaneously low power, low cost high data-rate wireless communications
- 7.5 Ghz of “free spectrum”
- Simple CMOS(Complementary Metal-Oxide-Semiconductor) transmitters at very low power
- “Moore’s Law Radio”

UWB Characteristics

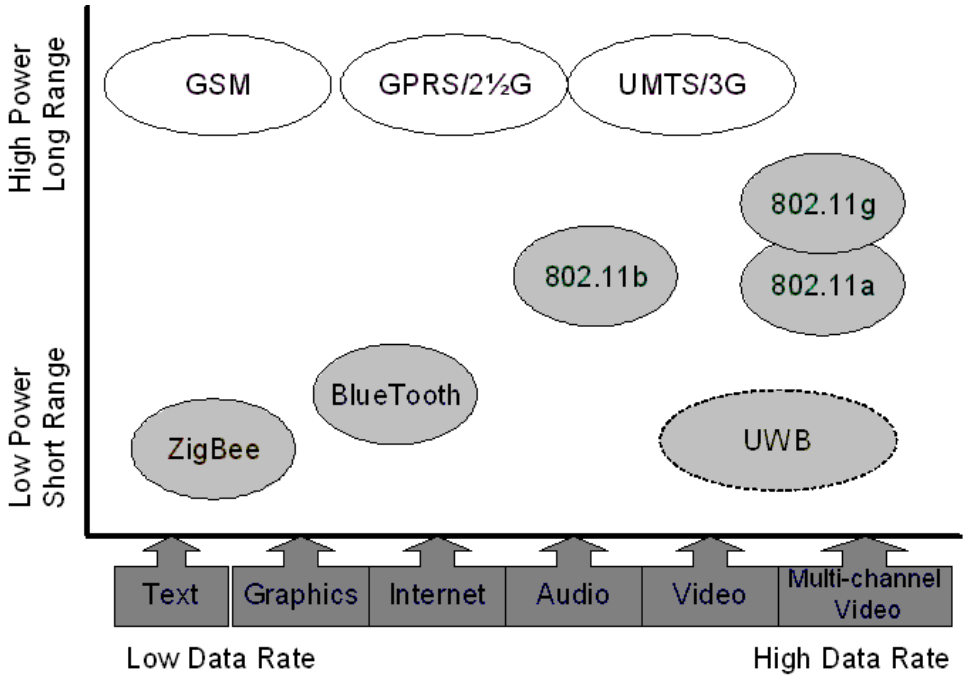
- Extremely low transmission energy
- Extremely difficult to intercept
- Multipath immunity to fading
- Follows Shannon’s channel capacity theorem

Advantages of UWB

- Does not use the traditional radio frequency carriers
- UWB is not line-of-sight
- Reduced Cost vs. Current Wireless Technology
- Channelization: Frequency sharing with other applications
- Transmission of data in large bursts
- Can trade-off throughput for distance

Comparison with other technologies

- Faster than Bluetooth, Wi Fi
- Data rate of 450Mbps instead of 1Mbps
- Complementary to existing radio technologies like 802.11



UWB Applications

- Replacing IEEE 1394 cables with wireless connectivity
- No. 1 High quality wireless video
- Accurate Ranging information
- Radar and Imaging

Challenges

- Interference with other licensed bands
- Tradeoffs with noise

Conclusion

- Well suited for high speed, short range WPAN
- Supports multimedia data rates, and offers inherent data security.
- Dominant player if FCC lightens the limitations

END